Physica A 451 (2016) 23–35

Contents lists available at ScienceDirect

Physica A

journal homepage: www.elsevier.com/locate/physa

Bank supervision using the Threshold-Minimum Dominating Set

Periklis Gogas, Theophilos Papadimitriou, Maria-Artemis Matthaiou *

Department of Economics, Democritus University of Thrace, Greece

HIGHLIGHTS

- We introduce a novel methodology, the T-MDS, for efficient and swift bank supervision.
- The T-MDS is more efficient than the MDS and the MST in economics (and other) networks.
- We identify the most representative subset of banks (T-MDS) yielding an overview of the whole banking network behavior.
- We support that these banks can act as "sensors" of an imminent bank crisis.
- The identified neighborhoods can be used as pathways of possible bank contagion.

ARTICLE INFO

Article history:
Received 13 May 2015
Received in revised form 31 October 2015
Available online 14 January 2016

Keywords:
Complex networks
Minimum Dominating Set
Banking supervision
Monitoring optimization

ABSTRACT

An optimized, healthy and stable banking system resilient to financial crises is a prerequisite for sustainable growth. Minimization of (a) the associated systemic risk and (b) the propagation of contagion in the case of a banking crisis are necessary conditions to achieve this goal. Central Banks are in charge of this significant undertaking via a close and detailed monitoring of the banking network. In this paper, we propose the use of an auxiliary supervision/monitoring system that is both efficient with respect to the required resources and can promptly identify a set of banks that are in distress so that immediate and appropriate action can be taken by the supervising authority. We use the network defined by the interrelations between banking institutions employing tools from Complex Networks theory for an efficient management of the entire banking network. In doing so, we introduce the Threshold Minimum Dominating Set (T-MDS). The T-MDS is used to identify the smallest and most efficient subset of banks that can be used as (a) sensors of distress of a manifesting banking crisis and (b) provide a path of possible contagion. We propose the use of this method as a supplementary monitoring tool in the arsenal of a Central Bank. Our dataset includes the 122 largest American banks in terms of their interbank loans. The empirical results show that when the T-MDS methodology is applied, we can have an efficient supervision of the whole banking network, by monitoring just a subset of 47 banks.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

The financial crisis of 2008 was initially manifested within the US and subsequently spread to a large part of Europe and the rest of the world. It practically demonstrated the known complex interrelations of financial institutions. Bank failures increase the systemic risk defined as the risk of destabilization of the whole system. In a worst case scenario bank crises

* Corresponding author.
E-mail address: maria.matthaiou7@gmail.com (M.-A. Matthaiou).

http://dx.doi.org/10.1016/j.physa.2015.12.149
0378-4371/© 2016 Elsevier B.V. All rights reserved.

may drive the entire network to a collapse if they lead to a set of simultaneous and cascading failures. The broader is the expansion of contagion, the higher is the systemic risk. Banking crises are associated with significant welfare costs and lead to negative deviations from the long-run output trend [1].

The Glass–Steagall Act in the US (Banking Act of 1933) imposed the distinction between commercial and investment banking. Several decades later, the Gramm–Leach–Bliley Act of 1999 waved this distinction as a result of the pressure from banking institutions that were adversely hit by the process of disintermediation. Disintermediation is defined as the practice of financing directly from capital markets without the intermediation of banks. That left banks with a decreasing share in total financing. After the Gramm–Leach–Bliley Act, banking institutions were free to engage in securities trading and insurance contracts increasing their market share. This newly gained freedom led to the rapid development by banks of new and diverse financial instruments. Banks became more competitive to the direct financing sources from capital markets in organized exchanges. Nonetheless, the liberalization of the banking sector led to a significant increase in the systemic risk as banks are since exposed to securities (investment) risk as well among other types of risk. Many analysts believe that this change may be one of the most significant causes of the 2008 financial crisis. The lack of separation between commercial and investment banking activities, allows banks to be involved in securities trading. The latter exposes depositors to higher risks. As a result, the Obama administration in 2010 introduced the Consumer Protection Act also known as the Dodd–Frank Act in an effort to optimize the supervision of the financial sector and to minimize the systemic risk. However, the provisions of the Dodd–Frank Act are far from the strict separation between commercial and investment banking imposed by Glass–Steagall and thus cannot eliminate market risk in the banking sector. Consequently, the major concern of regulators lies in their ability to promptly locate episodes of bank distress in order to implement the necessary policies for the minimization of contagion and defuse potential financial crises. Thus, an effective and continuous monitoring of financial institutions is necessary for maintaining a healthy and stable banking system. Moreover, a stricter supervision and regulation of the banking system is also required by the new Basel Accord (Basel III) from the Basel Committee on Banking Supervision that commenced implementation within the US in 2013 and will complete in 2018.

Within this modern regulatory framework, supervising authorities play a crucial role in (a) prudent supervision before a crisis is realized and (b) defusing the effects of possible crises and limit the effects of contagion. An important issue, apart from finding the appropriate monitoring tools, is the appointment of such a regulatory authority. Blinder [2] quotes a number of arguments for setting a single authority, the Fed, as supervisor: (a) credibility, (b) creation of economies of scale, and (c) pursuance of financial stability. Boyer and Ponse [3] state that a single supervisory authority preserves a more efficient supervision of the banking system. Moreover, Rochet and Vives [4] points out that a single regulatory authority can create and take advantage of the economies of scale between Lender of Last Resort (LOLR) facility, monetary policy and supervision. In the case of the Eurozone, EU leaders decided in October 2012 to assign the supervision of the whole European banking network to a single authority, the ECB, on the grounds that supervision of the entire network by a single authority can result in a prompt and efficient intervention. Additionally, the credibility of the ECB will facilitate the task of the European Stability Mechanism (ESM) to raise the necessary funds from the international markets for possible bailouts. Considering the volume of banks to be monitored both in the US and the EU, approximately 3000 and 6000 respectively, this will prove to be a tedious and demanding task.

In this paper, we propose an additional monitoring tool which may serve as an extra layer or an auxiliary monitoring system for banking supervision. We try to identify a reduced version of the initial network which: (a) contains an adequate amount of the total information and (b) is easier to control and analyze. We refer to this as the *representation goal*. The information needed for a successful representation depends on the application and is controlled by a threshold level. We investigate the interrelations between banking institutions using tools from Complex Networks theory and more specifically Graph theory. In the empirical section, we explore and analyze the interrelations between banking institutions based on Pearson's cross-correlation metric. The construction of networks based on the cross-correlation of an underlying variable, is a widely used technique in economics. First, Mantegna [5] constructed a network based on stock price correlations. This idea was followed by a number of researchers and some closely related papers are by Refs. Onnela et al. [6], Tumminello et al. [7], Sandoval [8], Lyocsa et al. [9], Tabak et al. [10], Kumar and Deo [11], and Bonanno et al. [12]. The studies examine economic networks that are constructed based on the correlation coefficient. The theory of Complex Networks provides proper tools to identify and explore the complex structure of today's banking institutions.

These studies are separated in two groups depending on the underlying network model: (a) random networks, and (b) real world networks. In their seminal paper regarding random networks Allen and Gale [13] investigated the cascading effect of a banking crisis on a network of regions or economic sectors. The authors showed that a complete interbank market (each region is connected to all other regions) or an incomplete interbank market with low degree of interconnectedness are resilient to a liquidity shock. Conversely, in the case of an incomplete interbank market with high degree of interconnectedness the liquidity shock may spread to the whole network. Other empirical studies examining random networks, (e.g. Gai and Kapadia [14] and Leitner [15]), reach the same conclusion. In a highly connected network, the probability of contagion is diminished as the losses of a failing institution can be more widely dispersed, and absorbed, by other institutions.

On the other hand the studies of, Angelini et al. [16], Iori et al. [39], Iori et al. [40], Fagiolo et al. [17], Nacher and Akutsu [41], Schiavo et al. [18], Boss et al. [38], and Inaoka et al. [19] analyze the topology of real-word networks and show that banking networks form scale-free and core–periphery structures, where a small number of hub banks have the most edges, while the rest (periphery) have just a few edges. These networks are resilient as far as the hubs, are robust to the

shock in the system. If hub-banks suffer a shock, it will be transmitted to the largest part of the network causing a systemic risk, unless these banks are backed by sufficient funds to stop the dispersion. On the contrary, a shock in a periphery node generates no problems to the stability of the system [18].

As banking networks have a complex structure, and in order to fully explore the roots of systemic risk, a number of papers concentrate on the topology of real-world economic and financial networks. Their goal is to identify the degree of banks' connectedness, the network's distribution and if possible, draw conclusions about the network's configuration during and after banking crises. Tabak et al. [20], introduce the directed clustering coefficient as a measure of systemic risk in complex networks. The results reveal that the network is not exposed to systemic risk. The clustering coefficient and domestic interest rates reveal negative correlation: as interest rates increase, the banks decrease their exposure to the system. Minoiu and Reyes [21], analyze the topology of the global banking network (GBN), formed of financial flows during and after periods of financial stress. The findings show that there is a number of structural breaks in the network, which indicates the waves of capital flows before and after crises. Network's centrality falls at the beginning and after a debt crisis, also the network is more stable in the second half of 2000 than in earlier periods. At last Thurner et al. [22], study whether the formation of a banking network has an aftereffect in an economy's health and conclude that well connected networks have fewer possibilities to default and that the safety of a network increases with the connectivity of the network.

Chan-Lau [23] uses a balance-sheet based model to investigate whether the risk of interbank exposure and the externalities with *too-connected-to-fail* institutions acted as a transmission mechanism triggered by a defaulted bank and resulted to the recent financial crisis. The results show that when the shocks in the network jeopardize banks' solvency, they can be characterized as sources of financial contagion worldwide. Plenty of studies use the balance-sheet approach to explore the different credit relations that exist between various banking systems. These studies test the propagation of contagion in a network resulting from the mutual claims between banks. Among them, Furfine [24] studies the US banking system, Upper and Worms [25] the German system, Wells [26] analyzes the UK interbank market, while Cocco et al. [27] the Portuguese one. The Swiss system is tested by Sheldon and Maurer [28] and Nagurney and Siokos [42] present evidence of contagion in the Belgian interbank market. The above studies yield the same conclusion: Banking systems are mainly resilient to shocks, while only a very small portion of banks will be affected by the failure of a single institution.

The closest related paper to ours is the one of Papadimitriou, Gogas and Tabak [29]. In this paper, by applying the Minimum Spanning Tree (MST), the authors identify the network's "core" banks and outline potential contagion paths in case of a bank distress. They suggest that core banks can be used as warning signals of an upcoming bank crisis. When a red flag is raised, the central bank must focus on the directly linked banks and take the necessary action for the minimization of systemic risk. The scope of our study is to reinforce and extend the optimal monitoring of the banking network using tools from Complex Networks theory. We propose the use of the novel *Threshold-Minimum Dominating Set* (T-MDS) as an auxiliary bank network monitoring system. This methodology has the advantage that (a) it does not impose any unnecessary or inappropriate restrictions in its algorithmic optimization solution as the MST does[1] and (b) by employing a thresholding step uses only the most relevant information for application in an economics network.

The filtering of edges is not new in Complex Networks methodologies. Onnela et al. [30,6] in papers treating stock market based networks, used the idea of keeping just a predetermined number of edges in the processing. Our thresholding idea comes closer to the study of Tse et al. [31], where a specific threshold is imposed in the correlation valued edges of the network.

The MDS has been proposed in the effort to solve the routing problem in mobile ad hoc wireless networking systems; see Refs. [32–34] and [43]. It consists of a system of mobile hosts connected by wireless links and it is extremely important in establishing communication in inhospitable terrains or battlefields. The problem in these unfriendly environments is that the standard solution of network flooding for the transmission of information is not an acceptable solution since (a) it creates high protocol overhead, (b) the transmitted information is often altered due to communication interference, multipath fading, or noise, and (c) the message may not even reach its destination if the network is sparse enough. In order to overcome the problems of the traditional routing protocols, a virtual backbone-based routing strategy has been introduced by Das and Bharghavan [35], and Johnson and Maltz [36]. The proposed solution is the creation of a subnetwork of routers (dominant nodes), forming the backbone of the system. All the information of the network is transmitted through the backbone and not through the whole network (as in the case of "flooding"). The success of this idea imposes that (i) every router is a backbone router or connected to at least one backbone router (so that every message can access the backbone to reach its destination), (ii) all routers in the backbone are connected and (iii) the backbone subnetwork is as small as possible (so that there is a minimum use of the network) [33]. MDS-based methods have been used in various studies for the identification of the backbone subnetwork.

Our approach differs from the backbone strategy in the sense that we do not require a subnetwork linking the dominant nodes (this is more the case of the MST-based methodologies), though we adopt the same principle that invokes the use of the MDS algorithm: we need every node in our network to be a dominant node or directly attached to a dominant node, and we need a small set of dominant nodes. T-MDS can reduce that amount of banks that need to be monitored by the Central Bank as it exploits the similarities of banks. Highly correlated banks make monitoring all of them unnecessary. Thus, since exhaustive supervision is associated with high costs and delays the T-MDS can reduce the cost of an efficient supervision and also possibly minimize the Central Bank's response time to the banking institutions in stress.

[1] The algorithms identifying the MST of a graph, impose that no closed loop is formed by the edges of the spanning tree.

We must also make clear here that we do not propose that the T-MDS should substitute current supervision schemes, but it may be used as an efficient early warning system. The ultimate goal is to provide an additional monitoring mechanism that can also act as a warning system of an upcoming bank distress. The proposed scheme locates the smallest subset of banks that best represent the entire banking network.

The rest of the paper is organized as follows. In Section 2 we provide a brief discussion on complex networks and we provide a detailed description of the T-MDS model. In Section 3 we present our dataset. In Section 4 we compare the T-MDS nodes, against the more central nodes according to specific centrality measures. Empirical results can be found in Section 5. The paper concludes in Section 6.

2. Proposed methodology

The main goal of this study is to provide a tool that may: (a) serve as an auxiliary monitoring tool of the entire banking network and (b) act as an early warning system of manifesting financial crises. These goals can be achieved through a small set of representative institutions.

To achieve this, we need to locate a subset of banks with two conflicting features: (a) small in size and yet (b) representative of the entire network. Moreover, the same subset has the ability to serve as a warning system for the regulators. When a red flag is raised by one of the identified representative banks the regulator can focus his/her attention on this bank's neighborhood and take the necessary measures to limit contagion and minimize systemic risk.

A Graph represents a collection of nodes that are interconnected. A node can represent anything: individuals, entities, computers, or as in our case banks. Two nodes are interconnected if there exists a link ("edge" in Graph Theory terminology) between them.

Definition 1. A **graph** G consists of a collection V of **nodes** and a collection of **edges** E, for which we write $G = (V, E)$. Each edge $e \in E$ is said to join two nodes. The structure can be extended by assigning a weight parameter to each edge, representing connections with numerical values.

We must define three key concepts for the understanding of our methodology:

- *Isolated node* is a node that is not connected to any other node in the network.
- *Interconnected node* defines the opposite term than the isolated node, a node that is connected at least with one other node in the network.
- *Dominant node* is a representative node for a neighborhood of nodes in the network.

When graph theory is applied in economics, the nodes represent the economic entities, the edges represent the links between nodes, and the weights represent the similarity measure of pairs of nodes on the tested variable. The depiction of nodes and their corresponding edges form a graph. In our study the nodes of the network represent the 122 largest American banks in terms of their interbank loans, while the weights measure the similarity of the interbank loans, more precisely we use a four quarter moving average of interbank loans (IL), as calculated by the cross-correlations. For the calculation of cross-correlations we use the Pearson coefficient $r_{i,j}$ using the following equation:

$$r_{i,j} = r_{j,i} \stackrel{\triangle}{=} \frac{\mathrm{COV}\left(IL_i, IL_j\right)}{\sqrt{\mathrm{VAR}\left(IL_i\right)\mathrm{VAR}\left(IL_j\right)}}. \tag{1}$$

The Pearson coefficient takes values from $[-1, 1]$. Values near the maximum indicate a strong positive correlation; values near the opposite extreme indicate a strong negative correlation. When the coefficient takes values close to zero, there is no correlation between the banks. We use a four quarter simple moving average of interbank loans to reduce possible noise in the form of short-run fluctuations in the interbank loans series, as the value of statement's day interbank lending from the financial report may not accurately represent the funding needs of the whole quarter. The moving average is calculated by taking the arithmetic mean of a given set of values. It is called "moving" because old values are dropped as the new data becomes available and this causes the average to move along the time scale.

In order to identify the smallest subset of banks that can best represent the whole network we will use the MDS. As a first step, to better comprehend the concept of the MDS we give the definition of a Dominating Set (DS).

Definition 2. A **dominating set** of a graph G is a subset of nodes $\mathrm{DS} \subseteq V$ such that every node $\notin \mathrm{DS}$ is joined to at least one member of the DS by some link [37].

The methodology for labeling a group of nodes as DS nodes is the following: Let us consider a network of n nodes. We create n binary variables $x_i,\ i = 1, 2, \ldots, n$ such that

$$x_i = \begin{cases} 0, & \text{if } i \notin \mathrm{DS} \\ 1, & \text{if } i \in \mathrm{DS} \end{cases} \tag{2}$$

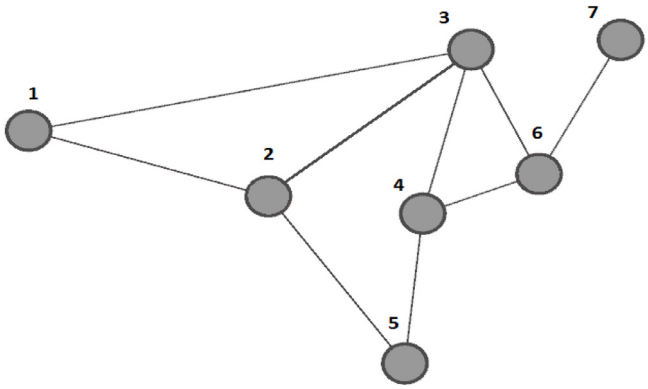


Fig. 1. A 7-node network.

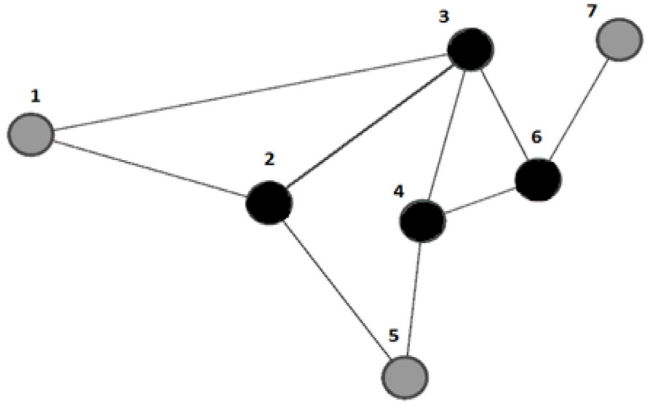


Fig. 2. In black we depict the nodes of a Dominating Set (DS) of the network.

to represent the nodes' membership to the DS. These variables may be represented in vector form as $\mathbf{x} = [x_1, x_2, \ldots, x_n]$. Definition 2 can be transformed to the following mathematical relation:

$$x_i + \sum_{j \in N(i)} x_j \geq 1, \quad i = 1, \ldots, n \tag{3}$$

where $N(i)$ represents the set of neighbors (adjacent nodes) of node i. The assumption is straightforward to follow: the node i can be (a) a node of the DS ($x_i = 1$) or (b) adjacent to at least a DS node ($\exists j \in N(i) : x_j = 1$).[2] In any case the l.h.s of the constraint is equal to or greater than 1. The imposed constraint preserves that every node of the network will be represented from the DS subset. Fig. 1 depicts the 7-nodes network.

In Fig. 2 we present one of the many DS that we can identify in the network of Fig. 1. Dominant are nodes 2, 3, 4, and 6. The nodes 1, 5, and 7 are represented through the dominant ones: node 1 is represented by dominant nodes 2 and 3, node 5 is represented by dominant nodes 2 and 4, and node 7 is represented by dominant node 6.

Definition 3. The Minimum Dominating Set (MDS) is the DS with the smallest cardinality.

More formally:

$$\min_{\mathbf{x}} \sum_{i=1}^{n} x_i. \tag{4}$$

To identify the MDS we must minimize Eq. (4) under the constraint of Eq. (3). Fig. 3 illustrates the MDS of the network in Fig. 1.

The MDS is the DS with the smallest cardinality. Fig. 3 depicts the MDS of the seven node example. Minimum Dominating Set is composed by nodes 2 and 6; nodes 1, 3, and 5 are represented by dominant node 2 and nodes 3, 4, and 7 are represented by dominant node 6. Finally, node 3 can be represented by both the dominant nodes of the network.

[2] This is not an exclusive disjunction: we may find nodes that verify both cases.

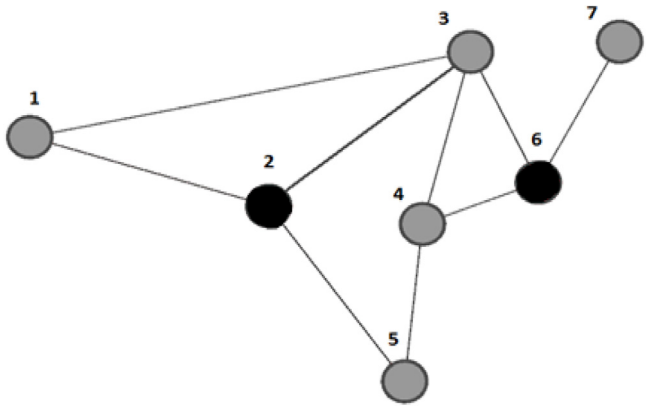


Fig. 3. The Minimum Dominating Set (MDS) is depicted with the black nodes.

The solution to the MDS problem may not be unique. Several subsets with the same minimum cardinality may represent the entire network. In the example of Fig. 3 the MDS is unique. The network regulator may have the option to choose between the alternative Dominating Sets. Another solution may be to estimate a weight on the nodes, and transform the summation in the minimization procedure into a weighted summation. This is an angle for future work on the T-MDS idea. The identification of the MDS is an NP-complete problem, and may yield a computationally demanding minimization process as we increase the size of the examined network. In an attempt to test the efficiency of the proposed methodology we applied the T-MDS method in a large sample dataset. In these experiments we were able to work with real data networks involving more than 4000 banks thresholded to 0.8, and more than 7000 banks thresholded to 0.9. These experiments showed that the proposed methodology can treat real banking networks comparable to the US or EU magnitude.

The cardinality of the dominant set depends on the connectivity of the network. The more connected is a network the fewer dominant nodes will be identified. The identification of the dominant nodes is achieved through a minimization procedure on the cardinality of the MDS (the integer programming problem in Eq. (4)). Thus, (a) the minimum number of nodes is an objective goal, and (b) the MDS nodes are identified directly as a set, and not sequentially. Why is this pertinent to our bank supervision goal? The denser is a network after the thresholding step, the more similar are the underlying banks as it is evident by the correlation. So a network with highly similar banks, can be supervised using a smaller set of dominant nodes/banks. Every dominant node represents a neighborhood of nodes. The importance is not the identity of the specific dominant bank but its high correlation to the other banks in the neighborhood that allows us to use it as a gauge for the behavior of the whole neighborhood. If two neighborhoods are overlapped the two dominant nodes may have some common features. Minimum number of dominant nodes in a general sense means that (a) the dominant nodes are as different as possible and (b) the neighborhoods are the less possibly overlapped. In fact ideally the neighborhoods should have empty intersection set.

Initially we start with a complete network of the banking system, i.e. a network where every pair of nodes is connected. However, not every edge of this network contains relevant information to our cause. Our interest lies only on the edges that represent a strong positive correlation. A low correlation edge corresponds to a pair of banks with low similarity. The information acquired by these edges is irrelevant or even misleading to our goal. Thus, we algorithmically impose the elimination from the initial network of all edges that represent a correlation below a threshold level. We must point that the thresholded network: (a) is a sparser network (i.e. a network with fewer edges than the initial complete one) and (b) it may have isolated nodes. Nevertheless the remaining interconnected nodes of the network are highly correlated (at least with one more node).

Definition 4. We call a **Threshold-Minimum Dominating Set** the two step methodology for the identification of the most representative nodes of a network, defined as:

Step 1. a Thresholding on the weights leading to the elimination of the low correlation edges and then
Step 2. the identification of the MDS nodes on the remaining network.

The imposition of a threshold in Step 1 of Definition 4 is an improvement over the standard MDS methodology as it allows us to take into account in the algorithmic optimization procedure for the identification of the Dominant nodes, only the meaningful (strong) bank interrelations.

It is important to clarify the term *representative nodes* used in Definition 4: we claim that every dominant node in T-MDS can act as a sensor for its directly linked neighborhood nodes, since they are connected with edges that survived the Thresholding step (i.e. they are highly correlated). The directly linked banking institutions are called neighborhoods that are represented by a Dominant node that can be used as a gauge for the monitoring of the neighborhood's banks.

So as we can see, the T-MDS method is an optimized tool as it overcomes the no loop restrictions that the MST method poses. The no loop restrictions result in the elimination of edges/links which are essential for the network as they connect

Table 1
Remaining edges' statistical significance tests (t-tests).

	Thresholding					
Threshold	0.4	0.5	0.6	0.68	0.75	0.8
Remaining edges	1390	1047	730	530	351	268
	Statistically significant edges					
$p = 0.05$	1390	1047	730	530	351	268
$p = 0.01$	1390	1047	730	530	351	268

Table 2
T-MDS metrics for various thresholds.

Strategy	1	2	3	4	5	6
Threshold	0.4	0.5	0.6	0.68	0.75	0.8
Dominant nodes	7	10	15	17	22	26
Isolated nodes	0	2	4	10	14	21
T-MDS nodes	7	12	19	27	36	47
Total edges	1390	1047	730	530	351	268

nodes with high similarity, and in the same time it retains edges that connect nodes with low similarity. Those are considered significant drawbacks that could influence the MST method's efficiency and are surpassed in the T-MDS method with the imposition of the Thresholding step. By definition every isolated node belongs to the T-MDS since it can be represented only by itself. In this case, the set of neighbors in Eq. (3) is the empty set, which means that (a) the summation part does not exist and consequently (b) the first part of the addition is equal to one. The rest of the nodes in the T-MDS are the interconnected nodes that represent their neighborhood: the dominant nodes. A distinction between isolated and dominant nodes should be done since they have some significant qualitative differences for the analysis of such networks. An isolated node represents a bank with a distinct idiosyncratic behavior as compared to the rest of the network; a dominant node represents a bank that has similar behavior with a group of banks and can be used as a distress gauge for them. According to the above, the overview of the whole network requires monitoring both kinds of T-MDS nodes (dominant and isolated), though it is essential to group and study them separately.

3. The data

We collected quarterly data for the largest 122 US banks, in terms of interbank loans. The sample period spans from 2002Q1 to 2012Q1. All data are publicly available from the financial statements of the corresponding institutions and they are collected from the Federal Deposit Insurance Corporation's (FDIC) website. The corresponding entry that we used is labeled as "loans to depository institutions and acceptances of other banks". We choose to use interbank loans as the variable of choice for calculating the network's correlations since these may play an important role in the transmission of bank distress (contagion) and thus they are directly interwoven with systemic risk. A banking institution that is faced with short-term liquidity shortage can borrow the necessary funds to cover its liquidity obligations through the interbank market. Therefore, banks are linked through their interbank loans and these interrelations increase the risk of contagion as a bank failure may trigger a domino effect contaminating large parts or even the entire network. Interbank loans are extremely short-term facilities with maturities ranging from overnight to a week and exhibit very high volatility and clustering. As such data taken from quarterly financial statements represent outstanding such loans only on the statement's date, they may not accurately represent interbank funding needs for the whole quarter. For this reason we use a four quarters moving average of interbank loans for normalization. Of course, for the analysis we may select additional alternative variables as well from the financial statements of these banks. These may be the total deposits, investment assets, provisions for non-performing loans, etc. The selection of the appropriate variable lies to the discretion of the bank supervisor. Depending on the scope of the analysis, alternative variables may be useful or necessary to be used.

4. Empirical results

We apply the T-MDS methodology to our sample of 122 US banks. In doing so, first, using Eq. (1) we calculate all cross-correlations and produce the adjacency matrix that creates the corresponding network. Then, we set the threshold level in order to eliminate all irrelevant to our analysis edges. In doing so, we tested a significant range of alternative threshold levels. For the edges that survived each threshold, we then used a t-test to evaluate their statistical significance ($H_0: \rho_{ij} = 0$). We report these results in Table 1.

According to Table 1, all edges that survive the thresholding procedure are statistically significant even at the 1% level of significance, for all threshold levels. Thus, only the edges with relevant information remain in the resulting network. Next, we identify the dominant and isolated T-MDS nodes. The results are presented in Table 2. The empirical results in Table 2 show that for the lowest employed threshold level ($t = 0.4$) the T-MDS is composed of just 7 nodes. At the opposite side of the threshold range, a threshold level of $t = 0.8$ identifies 47 T-MDS nodes, with 21 of them being isolated nodes. According to this, the interconnected nodes in the thresholded network are 101, and they can be monitored by the 26 dominant nodes.

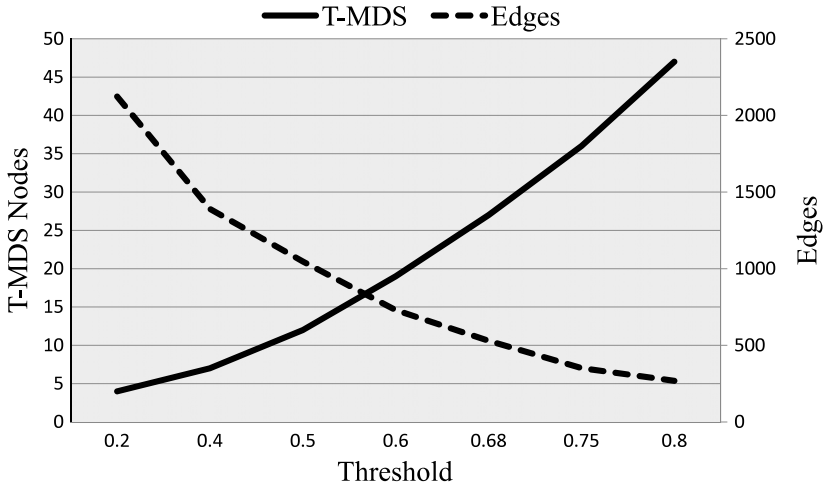


Fig. 4. The remaining network edges after the imposition of the threshold and the T-MDS nodes (connected and isolated nodes) for various threshold levels.

As it is evident from Table 2, the T-MDS size and the total number of remaining edges are inversely related. As the threshold is increased, the T-MDS nodes increase as well, while the edges decrease. Fig. 4 illustrates this observation.

The T-MDS is composed by the dominant and the isolated nodes. The inversely proportional relation with the threshold level applies in both classes of nodes.

- As the threshold level is increased by the regulator, more edges do not survive the threshold, eventually yielding nodes without any edges (in our example the isolated nodes appear once the threshold level reaches 0.5). The higher the threshold, the more nodes become isolated.
- Even if the nodes are not isolated, increasing the threshold level reduces the total number of edges resulting in the formation of smaller neighborhoods (since fewer nodes remain interconnected). The neighborhood concept appears in Eq. (3): the basic DS assumption, dictating that every interconnected node is either a DS node or directly linked to a DS node—at least one of the two cases is always true. As the neighborhood gets smaller, it is more probable that the interconnected node in question is not directly connected with a DS node, which in turn means that the node is a dominant node (and T-MDS node respectively). The higher the threshold, the more nodes become dominant nodes.

Figs. 5 and 6 depict the resulting thresholded network after imposing a threshold of 0.4 and 0.8 respectively. We choose two extreme threshold values in order to highlight the radical way in which the network's structure is affected. In both cases we have an overview of the whole banking network, but there is a sizeable difference in the number of connected nodes. In both figures the dominant nodes are depicted with dark color, while the rest of the nodes are white. The isolated nodes are distinct, as they have no edges.

By imposing a low correlation threshold, a smaller set is identified as the T-MDS. However, low threshold levels allow small correlations to survive the Step 1 of Definition 4. We should highlight here that the correlation metric is used to measure the similarity between banks. Thus, although a low threshold yields fewer nodes to monitor, this comes at the cost of low average similarity between nodes and this decreases the accuracy and increases the uncertainty of the supervision system. High thresholds result in the exactly opposite effect. Consequently, the regulator needs to tune the network using a threshold that is (a) high enough to maintain only the reliable interbank relations, but (b) not too high to yield a fragmented and sparse network that is impractical.

5. Comparison with centrality metrics

We claim that the T-MDS nodes are the most informative ones in the effort to monitor the entire network. To illustrate this, we compare the T-MDS subset to the nodes with the highest values according to three centrality metrics (Node Centrality, Eigenvalue Centrality, and Betweenness Centrality). Network centrality is often used to identify the most important nodes in the sense of cohesiveness. Thus, the notions of centrality and dominance are closely related.

In Tables 3–5 we present the centrality metrics for the ten banks with the highest values in descending order.

In Table 3 we can see that 6 out of the 10 banks with highest values in betweenness centrality are also dominant banks. Table 4 shows that only 2 out of the 10 nodes with the highest values in node degree are dominant banks. Finally, in Table 5 only 1 of the nodes with the highest Eigenvector Centrality is dominant. So this simple simulation shows that the betweenness centrality is a measure that seems to show some correlation with the T-MDS nodes.

Furthermore, we created a simple algorithmic procedure (similar to the one introduced in Ref. [29]) for each of the three centrality metrics to test the efficiency of these metrics with respect to the T-MDS in representing the entire network. We

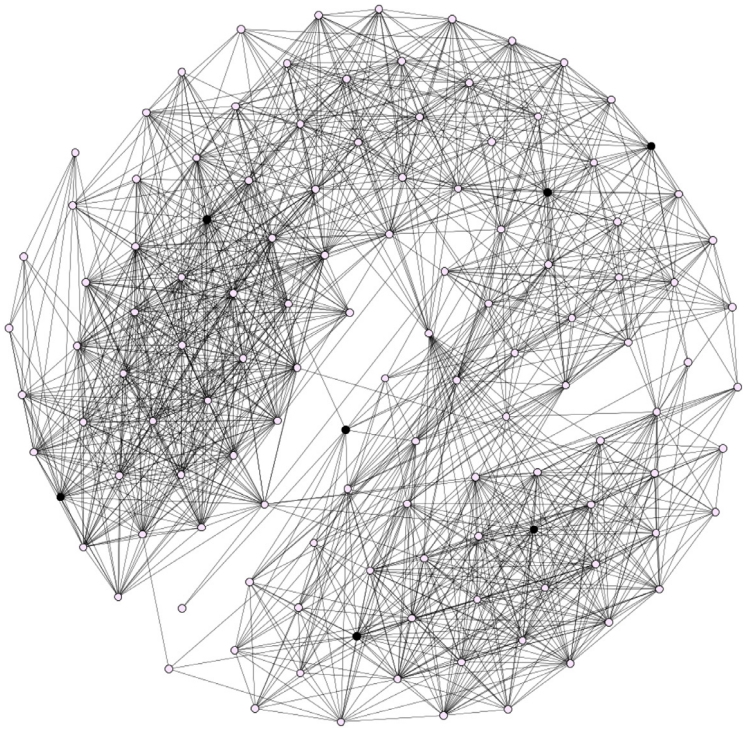


Fig. 5. Network after the imposition of a threshold $= 0.4$, black nodes are the dominant nodes, white nodes represent the rest nodes of the network.

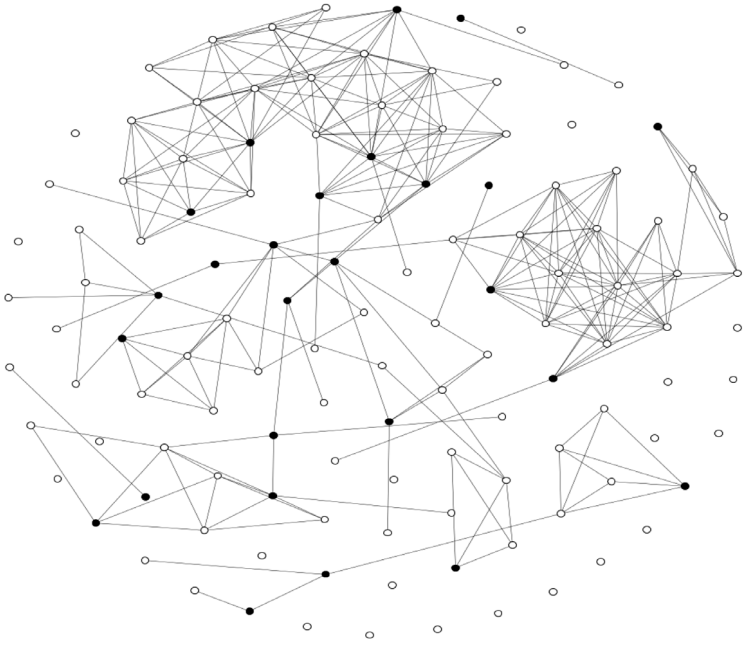


Fig. 6. Network after the imposition of a threshold $= 0.8$, black nodes are the dominant nodes, white nodes represent the rest nodes of the network.

consider the interconnected part of the network after the elimination of the edges below the threshold (threshold 0.8) involving 101 nodes sorted according to the centrality metrics.

Algorithm 1.

Step 1. Repeat until all the nodes are monitored
Step 2. Find node *i* with the highest centrality value

Table 3
Top 10 banks with respect to betweenness centrality.

Institution	Betweenness centrality
US Bank[a]	834.0
United Bank	754.8
Renasant Bank[a]	718.0
Bank of Nevada[a]	593.7
First Citizens Bank and Trust Company, Inc.[a]	579.0
Bank of India	383.1
State Bank of India	343.5
MERCANTIL COMMERCEBANK[a]	281.8
Heartland Bank and Trust Company[a]	259.0
Zions First National Bank	254.5

[a] Denotes a T-MDS Dominant bank.

Table 4
Top 10 banks with respect to Node centrality.

Institution	Node centrality
The Huntington National Bank	14
MERCANTIL COMMERCEBANK[a]	14
Bank of Baroda	14
Bank of India	14
JP Morgan Chase bank	13
California Bank & Trust	13
Cathay Bank[a]	13
State Bank of India	13
Manufacturers and Traders Trust Company	12
UMB Bank	12

[a] Denotes a T-MDS Dominant bank.

Table 5
Top 10 with respect to Eigenvector centrality.

Institution	Eigenvector centrality
Bank of Baroda	1.000
MERCANTIL COMMERCEBANK[a]	0.952
JP Morgan Chase bank	0.945
California Bank & Trust	0.942
Bank of India	0.938
The Huntington National Bank	0.894
State Bank of India	0.889
UMB Bank	0.836
First Bank	0.836
Manufacturers and Traders Trust Company	0.816

[a] Denotes a T-MDS Dominant bank.

Table 6
Network's supervision with various metrics.

Threshold: 0.8 Connected banks: 101	
T-MDS	26
Betweenness centrality	33
Eigenvector centrality	35
Node degree	37

Step 3. If neighborhood of i is not fully monitored
Step 3a. Adjust the neighboring nodes' degrees.
Step 3b. Update the monitored nodes
- the node i is monitored
- the neighborhood of i is monitored

The algorithm identifies sequentially the nodes with highest centrality values that can be used to monitor a subset of the network. It ends when the whole network is monitored. The results are presented in Table 6.

According to these results, the T-MDS is the smallest representation set of the four identified ones.

Table A.1
The network's T-MDS nodes in the case of threshold = 0.8.

1. JP Morgan Chase bank		62. Emigrant Bank	**Isolated**
2. Bank of America'		63. Apple Bank for Savings	
3. Citibank		64. Trustmark National Bank	**Isolated**
4. Wells Fargo Bank	**Isolated**	65. MB Financial Bank	**Dominant**
5. US Bank	**Dominant**	66. Old National Bank	
6. PNC Bank	**Isolated**	67. Citizens Bank	**Dominant**
7. The Bank of New York Mellon		68. Fulton Bank	
8. Capital One		69. Banco Popular North America	
9. State Street Bank and Trust Company		70. National Penn Bank	**Dominant**
10. TD Bank		71. First Citizens Bank and Trust Company, Inc.	**Dominant**
11. SunTrust Bank	**Isolated**	72. First Midwest Bank	
12. FIA Card Services		73. First Interstate Bank	
13. Regions Bank	**Dominant**	74. Bank Hapoalim B.M.	
14. Fifth Third Bank	**Isolated**	75. State Bank of India	
15. The Northern Trust Company		76. MERCANTIL COMMERCEBANK	**Dominant**
16. BMO Harris Bank		77. United Community Bank	
17. KeyBank		78. Hancock Bank	
18. Manufacturers and Traders Trust Company		79. PlainsCapital Bank	
19. Capital One Bank (USA)		80. The Park National Bank	
20. Morgan Stanley Bank		81. First Financial Bank	**Isolated**
21. Discover Bank		82. First Bank	
22. Compass Bank		83. Wesbanco Bank, Inc.	
23. Comerica Bank	**Dominant**	84. BancFirst	**Dominant**
24. Bank of the West		85. First Commonwealth Bank	
25. Deutsche Bank Trust Company Americas	**Isolated**	86. Chemical Bank	
26. The Huntington National Bank		87. Cole Taylor Bank	**Isolated**
27. Wells Fargo Bank South Central		88. Safra National Bank of New York	
28. American Express Centurion Bank	**Isolated**	89. BBCN Bank	
29. Citizens Bank of Pennsylvania	**Dominant**	90. Pacific Western Bank	
30. Bank of China		91. Bank Leumi USA	
31. GE Capital Bank		92. Bank of Baroda	
32. City National Bank		93. Pinnacle Bank	
33. BOKF	**Isolated**	94. Beneficial Mutual Savings Bank	
34. Synovus Bank	**Isolated**	95. United Bank	
35. First Tennessee Bank		96. Columbia State Bank	**Dominant**
36. Associated Bank	**Dominant**	97. City National Bank of Florida	**Dominant**
37. Frost Bank	**Dominant**	98. Bank of India	
38. East West Bank		99. Intrust Bank	
39. Commerce Bank	**Isolated**	100. First Security Bank	
40. Silicon Valley Bank		101. First Merchants Bank	**Dominant**
41. Susquehanna Bank		102. Renasant Bank	**Dominant**
42. Zions First National Bank		103. Mizuho Corporate Bank (USA)	
43. BNY Mellon		104. Nevada State Bank	
44. Wells Fargo Bank Northwest	**Isolated**	105. Great Southern Bank	
45. First Hawaiian Bank	**Dominant**	106. Johnson Bank	**Isolated**
46. Valley National Bank		107. Sabadell United Bank	**Dominant**
47. TD Bank USA		108. The Bancorp Bank	**Isolated**
48. UMB Bank		109. Amarillo National Bank	
49. Prosperity Bank		110. Busey Bank	
50. First National Bank of Omaha	**Isolated**	111. EagleBank	
51. The PrivateBank and Trust Company	**Isolated**	112. Customers Bank	**Isolated**
52. Bank of Hawaii		113. Ocean Bank	
53. BancorpSouth Bank		114. Community Trust Bank	**Dominant**
54. Whitney Bank		115. Lake City Bank	
55. Arvest Bank		116. Bank of Nevada	**Dominant**
56. Firstmerit Bank	**Isolated**	117. RBC Bank, (Georgia)	**Dominant**
57. Amegy Bank	**Isolated**	118. Heartland Bank and Trust Company	**Dominant**
58. Umpqua Bank	**Dominant**	119. City National Bank of West Virginia	**Dominant**
59. California Bank & Trust		120. Hanmi Bank	**Dominant**
60. Cathay Bank	**Dominant**	121. First American Bank	
61. Israel Discount Bank of New York		122. Liberty Bank of Arkansas	
62. Emigrant Bank			

6. Conclusions

In this paper we investigated the application of tools derived from Graph Theory for the optimum supervision of complex banking networks. Moreover, we introduce a novel methodology for the identification of the subset of banks that can act as gauges of their respective neighborhoods, the Threshold-Minimum Dominating Set (T-MDS). The T-MDS is superior to other Complex Network methodologies previously used to analyze economic networks as: (a) it does not impose any unnecessary

or inappropriate restrictions in its algorithmic optimization solution and (b) by introducing a thresholding step it takes into account only the most relevant information for application in an economics network. The methodology identifies the T-MDS set, i.e. the set of *Dominant* and *Isolated* nodes that can effectively represent the entire network. The network's supervisor, the Central Bank, can monitor the T-MDS nodes and exploit the information provided by these representative banks in the effort to efficiently monitor the system and swiftly identify and intercept a possible manifesting banking crisis. We apply the T-MDS concept and explored its monitoring efficiency in a network of the 122 US largest banks in terms of their interbank loans that was used as the key network variable. Various threshold levels were employed in the effort to evaluate the efficiency and sensitivity of the proposed methodology in supervising a complex banking network. The results show that using the T-MDS methodology the managing of a complex banking network can be optimized in terms of efficiency. The benefits of such optimization increase with the size of the network.

Acknowledgments

This research has been co-financed by the European Union (European Social Fund (ESF)) and Greek national funds through the Operational Program 'Education and Lifelong Learning' of the National Strategic Reference Framework (NSRF)—Research Funding Program: THALES. Investing in knowledge society through the European Social Fund (MIS 380292).

Appendix

Table A.1 presents the 122 examined banking institutions. The table also displays the results of the T-MDS method (Dominant and Isolated banks), after the imposition of a 0.8 threshold.

References

[1] G. Hoggarth, R. Reis, V. Saporta, Costs of banking system instability: some empirical evidence, J. Bank. Finance 26 (5) (2002) 825–855.
[2] A.S. Blinder, How central should the Central Bank be?, J. Econ. Lit. 48 (1) (2010) 123–133.
[3] P.C. Boyer, J. Ponce, Regulatory capture and banking supervision reform, J. Financ. Stabil. 8 (3) (2012) 206–217.
[4] J.C. Rochet, X. Vives, Coordination failures and the lender of last resort: Was Bagehot right after all?, J. Eur. Assoc. 2 (6) (2004) 1116–1147.
[5] R.N. Mantegna, Hierarchical structure in financial markets, Eur. Phys. J. B 11 (1999) 193–197.
[6] J.-P. Onnela, K. Kaski, J. Kertesz, Clustering and information in correlation based financial networks, Eur. Phys. J. B 38 (2004) 353–362.
[7] M. Tumminello, T. Di Matteo, T. Aste, R.N. Mantegna, Correlation based networks of equity returns sampled at different time horizons, Eur. Phys. J. B 55 (2007) 209–217.
[8] L. Sandoval, Pruning a minimum spanning tree, Physica A 391 (2012) 2678–2711.
[9] S. Lyocsa, T. Vyrost, E. Baumohl, Stock market networks: The dynamic conditional correlation approach, Physica A 391 (2012) 4147–4159.
[10] B.M. Tabak, T.R. Serra, D.O. Cajueiro, Topological properties of stock market networks: The case of Brazil, Physica A 389 (2010) 3240–3249.
[11] S. Kumar, N. Deo, Correlation and network analysis of global financial indicies, Phys. Rev. E 86 (2012) 026101(1)–026101 (8).
[12] G. Bonanno, G. Caldarelli, F. Lillo, S. Micciche, N. Vandewalle, R.N. Mantegna, Networks of equities in financial markets, Eur. Phys. J. B 38 (2004) 363–371.
[13] F. Allen, D. Gale, Financial contagion, J. Polit. Econ. 108 (1) (2000) 1–33.
[14] P. Gai, S. Kapadia, Contagion in Financial Networks, Bank of England, WP No. 383, 2010.
[15] Y. Leitner, Financial networks: contagion, commitment and private sector bailouts, J. Finance 60 (6) (2005) 2925–2953.
[16] P. Angelini, G. Maresca, D. Russo, Systemic risk in the netting system, J. Bank. Finance 20 (5) (1996) 853–868.
[17] G. Fagiolo, J. Reyes, S. Schiavo, The World-Trade web: Topological properties, dynamics, and evolution, Phys. Soc. 2 (2008) 1–44.
[18] S. Schiavo, J. Reyes, G. Fagiolo, International trade and financial integration: a Weighted network analysis, No 2007–11, OFCE, 2008.
[19] H. Inaoka, Hideki Takayasu, T. Shimizu, T. Ninomiya, K. Taniguchi, Self-similarity of banking network, Physica A 339 (3) (2004) 621–634.
[20] B.M. Tabak, M. Takami, J.M.C. Rocha, D.O. Cajuero, Directed clustering coefficient as a measure of systemic risk in complex networks, Physica A 394 (2014) 211–216.
[21] C. Minoiu, J.A. Reyes, A networks analysis of global banking: 1978–2009, International Monetary Fund, 2011, WP/11/74.
[22] S. Thurner, R. Hanel, S. Pichler, Risk trading, network topology and banking regulation, Quant. Finance 3 (4) (2003) 306–319.
[23] J.A. Chan-Lau, Balance Sheet Network Analysis of Too-Connected-to-Fail Risk in Global and Domestic Banking Systems, Vol. 10, International Monetary Fund, 2010.
[24] C. Furfine, Interbank exposures: Quantifying the risk of contagion, J. Money Credit Bank. 35 (1) (2003) 111–128.
[25] C. Upper, A. Worms, Estimating Bilateral exposures in the german interbank market: Is there a danger of contagion? Eur. Econ. Rev. 48 (3) (2004) 827–849.
[26] S. Wells, Financial Interlinkages in the UK Interbank Market and the Risk of Contagion, Working Paper No. 230 (Bank of England) 2004.
[27] J.F. Cocco, F.J. Gomes, N.C. Martins, Lending relationships in the interbank market, 2003, http://ssrn.com/abstract=568704i.
[28] G. Sheldon, M. Maurer, Interbank lending and systemic risk: An empirical analysis for Switzerland, Rev. Suisse Econom. Politique Statist. 134 (1998) 685–704.
[29] T. Papadimitriou, P. Gogas, B.M. Tabak, Complex networks and banking systems supervision, Physica A 392 (19) (2013) 4429–4434.
[30] J.-P. Onnela, A. Chakraborti, K. Kaski, J. Kertesz, A. Kanto, Asset trees and asset graphs in financial markets, Phys. Scr. T106 (2003) 48–54.
[31] C.K. Tse, J. Liu, F.C.M. Lau, A network perspective of the stock market, J. Empir. Finance 17 (4) (2010) 659–667.
[32] J. Wu, H. Li, A Dominating-Set-Based routing scheme in Ad Hoc wireless networks, Telecommun. Syst. 18 (1–3) (2001) 13–36.
[33] X. Cheng, X. Huang, D. Li, W. Wu, D. Du, A polynomial-time approximation scheme for the minimum-connected dominating set in ad hoc wireless networks, Networks 42 (4) (2003) 202–208.
[34] M. Yavad, V. Rishiwal, G. Arora, S. Makka, Modified minimum connected dominating set formation for wireless Adhoc networks, J. Comput. 1 (1) (2009) 200–203.
[35] B. Das, V. Bharghavan, Routing in Ad-Hoc networks using minimum connected dominating set, Communications 1 (1997) 376–380.
[36] D.B. Johnson, D.A. Maltz, Dynamic Source Routing in Ad Hoc Wireless Networks, Springer, USA, 1996.
[37] C. Shen, T. Li, Multi-documant summarization via the minimum dominating set, in: Proceedings of the 23rd International Conference on Computational Linguistics, Association for Computational Linguistics, 2010, pp. 984–992.
[38] M. Boss, H. Elsinger, M. Summer, S. Thurner, Network topology of the interbank market, Quant. Finance 4 (6) (2004) 677–684.
[39] G. Iori, S. Jafarey, F.G. Padilha, Systemic risk on the interbank market, J. Econ. Behav. Organ. 61 (4) (2006) 525–542.

[40] G. Iori, G.D. Masi, O.V. Precup, G. Gabbi, G. Caldarelli, A network analysis of the Italian overnight money market, J. Econom. Dynam. Control 32 (1) (2008) 259–278.
[41] J.C. Nacher, T. Akutsu, Dominating scale-free networks with variable scaling exponent: Heterogeneous networks are not difficult to control, New J. Phys. 14 (7) (2012) 1–24.
[42] A. Nagurney, S. Siokos, Financial Networks—Statics and Dynamics, Springer, Heidelberg, Germany, 1997.
[43] J. Schleich, H. Thi, P. Bouvry, Solving the minimum M-dominating set problem by a continuous optimization approach based on DC programming and DCA, J. Comb. Optim. 24 (4) (2011) 397–412.